

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Samarium Cobalt powder

**Molecular Formula** approx. ratio: Sm:Cu:Fe:Zr:Co; 25.5:5:14:3:52.5 wt  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>12608</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable solids Category 2

Substances/mixtures which, in contact with water, emit flammable gases Category 1

#### Health hazards

|                                             |             |
|---------------------------------------------|-------------|
| Acute Oral Toxicity                         | Category 3  |
| Acute Inhalation Toxicity - Dusts and Mists | Category 3  |
| Serious Eye Damage/Eye Irritation           | Category 2  |
| Respiratory Sensitization                   | Category 1  |
| Skin Sensitization                          | Category 1  |
| Germ Cell Mutagenicity                      | Category 1B |
| Carcinogenicity                             | Category 1B |
| Reproductive Toxicity                       | Category 1B |

#### Environmental hazards

Chronic aquatic toxicity Category 2

### Label Elements

**Signal Word****Danger****Hazard Statements**

H228 - Flammable solid  
 H260 - In contact with water releases flammable gases which may ignite spontaneously  
 H317 - May cause an allergic skin reaction  
 H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
 H350 - May cause cancer  
 H360 - May damage fertility or the unborn child  
 H319 - Causes serious eye irritation  
 H340 - May cause genetic defects  
 H411 - Toxic to aquatic life with long lasting effects  
 H301 + H331 - Toxic if swallowed or if inhaled

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P223 - Do not allow contact with water  
 P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
 P240 - Ground and bond container and receiving equipment  
 P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
 P272 - Contaminated work clothing should not be allowed out of the workplace  
 P280 - Wear protective gloves  
 P284 - In case of inadequate ventilation wear respiratory protection  
 P273 - Avoid release to the environment

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention  
 P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P362 + P364 - Take off contaminated clothing and wash it before reuse  
 P391 - Collect spillage

**Storage**

P402 + P404 - Store in a dry place. Store in a closed container

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

## Section 3 - Composition and Information on Ingredients

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Cobalt    | 7440-48-4 | 52.5     |
| Samarium  | 7440-19-9 | 25.5     |
| Iron      | 7439-89-6 | 14.0     |
| Copper    | 7440-50-8 | 5.0      |
| Zirconium | 7440-67-7 | 3        |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                                          |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                      |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                                                    |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                 |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                                               |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

approved class D extinguishers. Do not use water or foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Copper oxides, Zirconium oxide, Iron oxides, Samarium oxide, Cobalt oxides, Hydrogen.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to

contaminate ground water system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep container tightly closed in a dry and well-ventilated place.

**Incompatible Materials**

Acids.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component | New Zealand WEL                                        | Australia                                              | ACGIH TLV                                              | The United Kingdom                                                                                                                         |
|-----------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Cobalt    | TWA: 0.02 mg/m <sup>3</sup><br>Skin                    | TWA: 0.05 mg/m <sup>3</sup>                            | TWA: 0.02 mg/m <sup>3</sup>                            | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens.                                                       |
| Copper    | TWA: 0.01 mg/m <sup>3</sup>                            | TWA: 1 mg/m <sup>3</sup><br>TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup>                             | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr |
| Zirconium | TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | STEL: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> |                                                                                                                                            |

**Biological limit values**

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

ACGIH - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component | New Zealand                                               | Australia | ACGIH - Biological Exposure Indices                                                      | United Kingdom |
|-----------|-----------------------------------------------------------|-----------|------------------------------------------------------------------------------------------|----------------|
| Cobalt    | 15 µg/L (urine) end of shift at end of work week (Cobalt) |           | 15 µg/L<br>Medium: urine<br>Time: end of shift at end of workweek<br>Determinant: Cobalt |                |

### Appropriate engineering controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Multi-purpose/ABEK Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
**Recommended half mask:-** Particle filtering: EN149:2001 (or AUS/NZ equivalent)  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                            |                          |
|----------------------------|--------------------------|
| <b>Physical State</b>      | Solid                    |
| <b>Appearance</b>          |                          |
| <b>Odor</b>                | No information available |
| <b>Odor Threshold</b>      | No data available        |
| <b>pH</b>                  | No information available |
| <b>Melting Point/Range</b> | No data available        |
| <b>Softening Point</b>     | No data available        |

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Water Solubility</b>                        | Insoluble in water       |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Cobalt                                         | 5                        |                                          |
| <b>Vapor Pressure</b>                          | 23 hPa @ 20 °C           |                                          |
| <b>Density / Specific Gravity</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Particle characteristics</b>                | No data available        |                                          |

**Other information**

|                                                                               |                                                              |
|-------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Molecular Formula</b>                                                      | approx. ratio: Sm:Cu:Fe:Zr:Co; 25.5:5:14:3:52.5 wt           |
| <b>Flammable solids</b>                                                       | Burning rate or burning time = > 5 minutes and <= 10 minutes |
| <b>Substances/mixtures which, in contact with water, emit flammable gases</b> | Emitted gas ignites spontaneously                            |
| <b>Evaporation Rate</b>                                                       | Not applicable - Solid                                       |

## Section 10 - Stability and Reactivity

|                                         |                                    |
|-----------------------------------------|------------------------------------|
| <b>Reactivity</b>                       | Yes                                |
| <b>Stability</b>                        | Air sensitive. Moisture sensitive. |
| <b>Sensitivity to Mechanical Impact</b> | No information available           |
| <b>Sensitivity to Static Discharge</b>  | No information available           |
| <b>Hazardous Polymerization</b>         | No information available.          |
| <b>Hazardous Reactions</b>              | None under normal processing.      |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks.           |
| <b>Incompatible Materials</b>           | Acids.                             |

**Hazardous Decomposition Products** Copper oxides. Zirconium oxide. Iron oxides. Samarium oxide. Cobalt oxides. Hydrogen.

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure****Product Information**

|                   |                                    |
|-------------------|------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure. |
| <b>Eyes</b>       | Avoid contact with eyes.           |
| <b>Skin</b>       | Avoid contact with skin.           |
| <b>Ingestion</b>  | May be harmful if swallowed.       |

**Numerical measures of toxicity****(a) acute toxicity;****Oral**

Based on available data, the classification criteria are not met

**Dermal**

No data available

**Inhalation**

No data available

**Toxicology data for the components**

| Component | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation              |
|-----------|---------------------------|---------------------------|------------------------------|
| Cobalt    | LD50 = 6171 mg/kg ( Rat ) |                           | LC50 < 0.05 mg/L ( Rat ) 4 h |
| Samarium  |                           | LD50 > 2000 mg/kg ( Rat ) |                              |
| Iron      | 7500 mg/kg ( Rat )        |                           |                              |
| Copper    |                           |                           | LC50 > 5.11 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;**

No data available

**(c) serious eye damage/irritation;**

No data available

**(d) respiratory or skin sensitization;****Respiratory**

Sub Category 1B

**Skin**

Category 1

**Sensitization**

No information available

**(e) germ cell mutagenicity;**

Category 2

**(f) carcinogenicity;**

Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | New Zealand          | Australia | New South Wales | Western Australia | IARC     | EU           | UK | Germany |
|-----------|----------------------|-----------|-----------------|-------------------|----------|--------------|----|---------|
| Cobalt    | Suspected carcinogen |           |                 |                   | Group 2A | Carc Cat. 1B |    | Cat. 2  |

**(g) reproductive toxicity;**

Category 1B

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

**Target Organs**

None known.

**(j) aspiration hazard;**

Not applicable

Solid

**Symptoms / effects, both acute and delayed**

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Section 12 - Ecological Information**

**Ecotoxicity****Aquatic ecotoxicity**

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                | Water Flea                                    | Freshwater Algae                                           | Microtox |
|-----------|--------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|----------|
| Cobalt    | LC50: > 100 mg/L, 96h static (Brachydanio rerio)                               |                                               |                                                            |          |
| Copper    | Onchorhynchys mykiss: LC50=0.15 mg/L 96h<br>Cuprinus carpio: LC50=0.8 mg/L 96h | EC50: = 0.03 mg/L, 48h Static (Daphnia magna) | 0.0426-0.0535 mg/L EC50 72 h<br>0.031-0.054 mg/L EC50 96 h |          |

**Terrestrial ecotoxicity**

There is no data for this product

**Persistence and Degradability**

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence**

Insoluble in water, May persist.

**Degradability**

Not relevant for inorganic substances.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

May have some potential to bioaccumulate Product has a high potential to bioconcentrate

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Cobalt    | 5       | No data available             |

**Mobility**

Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

**Other adverse effects****Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods****Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be



landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

| Component                    | Hazchem Code                |
|------------------------------|-----------------------------|
| Zirconium<br>7440-67-7 ( 3 ) | 1Z<br>1Y<br>4Y<br>3Y<br>3YE |

### NZS 5433:2020

**UN-No** UN3089  
**Proper Shipping Name** Metal powder, flammable, n.o.s.  
**Technical Shipping Name** (Samarium, Cobalt powder)  
**Hazard Class** 4.1  
**Packing Group** III

### IATA

**UN-No** UN3089  
**Proper Shipping Name** Metal powder, flammable, n.o.s.  
**Technical Shipping Name** (Samarium, Cobalt powder)  
**Hazard Class** 4.1  
**Packing Group** III

### IMDG/IMO

**UN-No** UN3089  
**Proper Shipping Name** Metal powder, flammable, n.o.s.  
**Technical Shipping Name** (Samarium, Cobalt powder)  
**Hazard Class** 4.1  
**Packing Group** III

| Component                   | IMDG Marine Pollutant                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------|
| Copper<br>7440-50-8 ( 5.0 ) | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below

| Component | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL)                    |
|-----------|------------------------------------|--------------------------------------|----------------------------------------|--------------------------------------------------------|
| Copper    |                                    |                                      |                                        | 0.0014 mg/L (Freshwater)<br>0.0013 mg/L (Marine water) |

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component | New Zealand          |
|-----------|----------------------|
| Cobalt    | Suspected carcinogen |

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                                                                   |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Copper    |                                                                                           |                                                                                          | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

**Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cobalt    | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |
| Copper    | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                                                                                       | -                                                                                                     |
| Zirconium | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                                                                                       | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Cobalt    | 7440-48-4 | X     | X    | -      | -      | -   | KE-06060 | X     | X    |
| Samarium  | 7440-19-9 | X     | X    | -      | -      | -   | KE-30854 | X     | X    |
| Iron      | 7439-89-6 | X     | X    | -      | -      | -   | KE-21059 | X     | X    |
| Copper    | 7440-50-8 | X     | X    | -      | -      | -   | KE-08896 | X     | X    |
| Zirconium | 7440-67-7 | X     | X    | -      | -      | -   | KE-35607 | X     | X    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-----------|-----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Cobalt    | 7440-48-4 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |
| Samarium  | 7440-19-9 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |

|           |           |   |        |   |   |   |   |   |
|-----------|-----------|---|--------|---|---|---|---|---|
| Iron      | 7439-89-6 | X | ACTIVE | X | - | X | - | X |
| Copper    | 7440-50-8 | X | ACTIVE | X | - | X | - | X |
| Zirconium | 7440-67-7 | X | ACTIVE | X | - | X | - | X |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

NZIoC - New Zealand Inventory of Chemicals

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

NZS 5433:2020 - Transport of Dangerous Goods on Land

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

AICS - Australian Inventory of Chemical Substances

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists

PNEC - Predicted No Effect Concentration

OECD - Organisation for Economic Co-operation and Development

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

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Revision Summary SDS sections updated 2 3

### Disclaimer

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**End of Safety Data Sheet**